Parametric Heatsink Conjugate Heat Transfer Simulation Report

OpenFOAM 13 · foamMultiRun (Buoyant, Multi-Region) · 3-Point Fin-Pitch Sweep
Parallel/Serial Verified · 16,000 Steady-State Iterations per Case

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July 2026

Parameter	Value
Solver	foamMultiRun (buoyant, 3-region CHT)
Regions	fluid (air); solid → heatsink (Al) + cpu (Si) cell zones
CPU heat dissipation	150 W
Fan	Delta FFB0812VH (quadratic curve, fanPressure BC)
Fan outlet velocity (converged)	3.87 m/s
Inlet air temperature	300 K (27°C)
Verification	Serial vs. 8-core parallel agree to 0.001 K (5 mm baseline)
Sweep	3 mm (24 fins) / 4 mm (18 fins) / 5 mm (15 fins)
Best CPU temperature	336.96 K (63.96°C) — 3 mm pitch, 24 fins
Worst CPU temperature	343.90 K (70.90°C) — 5 mm pitch, 15 fins
Temperature reduction (5→3 mm)	6.94 K

1. Project Overview

This project presents a steady-state Conjugate Heat Transfer (CHT) parametric fin-pitch sweep for a CPU and fin-array heatsink inside a fan-driven duct, using OpenFOAM 13's foamMultiRun multi-region driver. Unlike a single fixed-geometry case, the mesh itself is generated parametrically: a Python script lays out the fin/channel geometry directly as a blockMeshDict from a fin pitch and fin count, so the design can be swept without rebuilding CAD geometry by hand. Three cases were run — 5 mm, 4 mm, and 3 mm fin pitch — to identify the thermal optimum for this fan/duct combination under 150 W CPU dissipation.

Key Features

- 3-region CHT: fluid air domain, aluminium heatsink, and silicon CPU die, coupled via coupledTemperature interfaces
- Fully parametric blockMeshDict generation (gen_cases_v2.py) — fin pitch and fin count are input parameters, not hand-edited geometry
- Fan modelled with a fanPressure boundary condition driven by a tabulated quadratic fan curve (Delta FFB0812VH), not a fixed-velocity inlet
- Volumetric heatSource fvModel applies 150 W directly to the CPU silicon cell zone
- Baseline 5 mm case cross-checked on 1 core (serial) and 8 MPI ranks — converged results agree to 0.001 K
- Three-point pitch sweep completed: 5 mm / 4 mm / 3 mm, each run to 16,000 steady-state iterations
- Diminishing-returns trend identified: 5→4 mm saves 5.84 K; 4→3 mm saves an additional 1.10 K

2. Geometry and Mesh

2.1 Parametric Geometry Generation

The heatsink/duct geometry is produced directly as an OpenFOAM blockMeshDict by gen_cases_v2.py, rather than imported from CAD. Given a fin pitch and fin count, the script lays out alternating fin / channel / side-gap strips across the 80 × 80 mm duct cross-section, splits those strips against the CPU footprint, and writes the resulting hex-block mesh with fan, wall, and region-zone tagging already built in. Each pitch case is a fresh blockMesh run from the same generator with different parameters — no manual geometry editing between cases.

Fixed geometry: duct cross-section 80 × 80 mm; fin thickness 1.5 mm; fin height 35 mm; base thickness 5 mm; CPU die 40 × 40 × 4.5 mm centred under the fin array; inlet and outlet plenum length 105 mm each (total duct length 290 mm). Side gaps are sized to fill the remaining duct width symmetrically for each pitch.

2.2 Per-Case Geometry

Case	Fin Pitch	Fins	Channel Width	Approx. Mesh Cells
p5_cpu	5 mm	15	3.5 mm	649,200
p4_cpu	4 mm	18	2.5 mm	~695,520
p3_cpu	3 mm	24	1.5 mm	~780,000

2.3 Mesh

Each case uses a block-structured hex mesh built with blockMesh and split into coupled regions using splitMeshRegions -cellZones. After splitting, topoSet -region cpu is run to recreate the allCpuCells cellZone (required by the heatSource fvModel) which is not preserved across a region split.

Region	Material	Role
fluid	Air (buoyant, k-ε)	Flow & heat transport
heatsink	Aluminium (k = 205 W/m·K, ρ = 2700 kg/m³)	Heat spreading / dissipation
cpu	Silicon (k = 150 W/m·K, ρ = 2329 kg/m³)	Heat source (150 W via heatSource fvModel)

3. Simulation Setup

3.1 Solver Configuration

Each case is driven by foamMultiRun, coupling one buoyant fluid region (solved with k-ε turbulence and a p_rgh pressure formulation) to a solid super-region containing the heatsink and cpu cell zones. Region interfaces use the coupledTemperature boundary condition at every fluid↔heatsink, fluid↔cpu, and heatsink↔cpu face. Heat is injected directly into the CPU silicon zone using a heatSource fvModel (cellZone allCpuCells, Q = 150 W).

Relaxation factors are set to p_rgh 0.4, U 0.6, h/k/ε 0.4 across all three cases — a conservative choice validated after the finer-pitch meshes were found to diverge at higher values (0.7–0.8). End time is 16,000 iterations, reduced from 20,000 after the 5 mm baseline confirmed convergence plateau by ~15,000 steps.

3.2 Fan Boundary Condition

Rather than a fixed-velocity inlet, the fan is modelled with a fanPressure boundary condition at the fan1_half0 patch, driven by a tabulated pressureVsQ.csv curve. The table is generated from the quadratic Delta FFB0812VH fan curve $dP = 73.99 - 1164.25 \cdot Q - 34615 \cdot Q^2$ [Pa, Q in m³/s], so the operating point of the fan against each duct's actual flow resistance is solved for — not assumed. As fin pitch tightens, flow resistance increases and the operating point shifts to lower flow rates.

3.3 Boundary Conditions Summary

Boundary	Field	Type	Value
fan1_half0 (inlet)	p_rgh	fanPressure	Delta FFB0812VH curve (table)
fan1_half0 / fan2_half0	U	pressureInletOutletVelocity	Solved from fan curve
base_bottom / top_wall / side_walls	U	noSlip	—
fluid↔heatsink, fluid↔cpu, heatsink↔cpu	T	coupledTemperature	Region-coupled
CPU cell zone	heatSource (fvModel)	heatSource	150 W

3.4 Verification: Serial vs. Parallel (5 mm Baseline)

The 5 mm-pitch baseline case was run twice to 20,000 iterations — once on a single core and once decomposed across 8 MPI ranks — to confirm that domain decomposition does not change the converged solution. Both runs converge to T_max = 343.900 K, agreeing to within 0.001 K.

4. Results

4.1 Parametric Sweep Summary

All three pitch cases ran to 16,000 steady-state iterations and reached convergence well before end time. CPU maximum temperature was monitored every iteration via the cpuTmax volFieldValue function object. The results show a clear monotonic decrease in CPU temperature as fin pitch decreases, with strongly diminishing returns below 4 mm.

Case	Fin Pitch	Fins	T_max (K)	T_max (°C)	ΔT vs. ambient	ΔT vs. 5 mm
p5_cpu	5 mm	15	343.90	70.90	43.90 K	— (baseline)
p4_cpu	4 mm	18	338.06	64.06	38.06 K	–5.84 K
p3_cpu	3 mm	24	336.96	63.96	36.96 K	–6.94 K

4.2 Convergence

Each case's T_max history (available in postProcessing/cpu/cpuTmax/*/volFieldValue.dat) shows a smooth monotonic climb from the initial condition and a clear plateau well before end time. The 5 mm baseline plateaus at ~15,000 iterations; the finer-pitch cases (larger meshes) follow the same pattern within the same 16,000-iteration budget. Residuals for U, h, e (both solid regions), and p_rgh are all below 1×10■■ at convergence.

4.3 Physical Interpretation

The dominant effect of tighter fin pitch is increased total fin surface area (15 → 18 → 24 fins), which enhances convective heat transfer and reduces CPU temperature. The opposing effect is increased channel flow resistance, which shifts the fan operating point to lower flow rates. Between 5 mm and 4 mm, the surface area gain dominates strongly (–5.84 K). Between 4 mm and 3 mm, the channel width narrows to 1.5 mm and the resistance penalty begins to erode the surface area benefit, yielding only –1.10 K. This strongly diminishing return indicates the thermal optimum for this fan/duct combination lies in the 3–4 mm range.

A further tightening to 2 mm (channel width 0.5 mm) was considered but not pursued: at that geometry the fan would operate at significantly reduced flow, and the net thermal benefit is expected to be negligible or negative — the 3 mm case already shows the trend turning.

4.4 Workflow

For each pitch case: gen_cases_v2.py (parametric blockMeshDict) → blockMesh (hex mesh with fin/channel/side-gap strips and fan/wall patches) → splitMeshRegions -cellZones (fluid | heatsink | cpu) → topoSet -region cpu (recreate allCpuCells cellZone) → foamMultiRun (coupled buoyant + conductive solve, 16,000 iterations) → postProcessing (cpuTmax, fan patch flow rate and velocity).

5. Status

All three cases of the fin-pitch sweep are complete:

Case	Fin Pitch	Fins	Status	T_max

p5_cpu (serial)	5 mm	15	Converged, 20,000 it.	343.90 K
p5_cpu (8-core parallel)	5 mm	15	Converged, 20,000 it.	343.90 K
p4_cpu	4 mm	18	Converged, 16,000 it.	338.06 K
p3_cpu	3 mm	24	Converged, 16,000 it.	336.96 K

6. Limitations

- Region coupling uses a simplified coupledTemperature boundary condition rather than a baffle-resistance formulation, so thermal interface material (TIM) contact resistance between the CPU die and heatsink base is not explicitly represented.
- Outer duct boundaries (base_bottom, top_wall, side_walls) use noSlip walls rather than a true open-atmosphere condition; the duct cross-section fully constrains the flow domain.
- The fan curve table is a quadratic fit to the published Delta FFB0812VH specification curve rather than measured operating data for this specific duct impedance.
- All three cases were run serially (single core) to avoid resource conflicts; wall-clock times of ~22 hours per case are acceptable for a portfolio study but would benefit from parallelisation in a production setting.
- The sweep covers three pitch points; the resolution is sufficient to identify the trend and the approximate optimal range (3–4 mm) but not to pinpoint a precise optimum to sub-millimetre accuracy.

7. Methodology Summary

- **Geometry:** Parametric blockMeshDict generation via gen_cases_v2.py (Python) — fin pitch and fin count as direct inputs; one fresh mesh per pitch case.
- **Mesh:** blockMesh hex mesh with fin/channel/side-gap strip decomposition; splitMeshRegions -cellZones for fluid/heatsink/cpu region split; topoSet -region cpu to recreate allCpuCells cellZone after each split.
- **Region Coupling:** coupledTemperature boundary condition at all fluid↔solid and solid↔solid interfaces.
- **Heat Source:** heatSource fvModel, cellZone allCpuCells, Q = 150 W.
- **Fan Modelling:** fanPressure boundary condition with tabulated quadratic Delta FFB0812VH curve (pressureVsQ.csv).
- **Solver:** foamMultiRun (buoyant fluid + conductive solid, k-ε turbulence), 16,000 steady-state iterations; relaxation p_rgh 0.4 / U 0.6 / h-k-ε 0.4.
- **Verification:** Baseline 5 mm case run serial and on 8 MPI ranks; converged T_max agrees to 0.001 K.
- **Post-processing:** cpuTmax volFieldValue function object; fan patch flow rate and velocity monitors.

Software Stack

Tool	Version	Purpose
OpenFOAM	13	CFD solver (foamMultiRun, buoyant multi-region CHT)
Python	3.x	Parametric blockMeshDict generation (gen_cases_v2.py)
Ubuntu (WSL2)	24.04	Operating system
OpenMPI	—	8-core parallel verification run (5 mm baseline)
reportlab	—	PDF report generation

8. Conclusion

A three-point parametric fin-pitch sweep (5 mm / 4 mm / 3 mm) was completed using OpenFOAM 13's foamMultiRun multi-region CHT driver with a fully parametric mesh generation pipeline. All three cases dissipate 150 W from a CPU die into a fan-driven duct with the Delta FFB0812VH fan modelled via its published pressure curve rather than a fixed flow rate.

The sweep demonstrates a clear and physically intuitive result: tightening fin pitch from 5 mm to 4 mm yields a strong 5.84 K reduction in peak CPU temperature (343.90 K → 338.06 K) driven by increased fin surface area. A further tightening to 3 mm adds only 1.10 K of improvement (338.06 K → 336.96 K), indicating that channel flow resistance is beginning to offset the surface area benefit. The thermal optimum for this fan/duct combination lies in the 3–4 mm range; further pitch reduction is expected to yield negligible or negative returns.

The full case files, including the parametric mesh generator and solver configuration, are available at github.com/samuelsimmons99/OpenFOAM-Portfolio.